MONASH University

# FIT1047 Applied Session Week 3

# BOOLEAN EQUATION AND KARNAUGH MAPS

## OBJECTIVES

- The purpose of this applied session is getting to know Boolean Expressions, and simplifications using K-maps.

## INSTRUCTIONS

- Use Logisim to draw circuits. Apply K-map to simplify equations.
- You may work in a small group.

## Activity 1: Boolean Equation to Circuit

Given a Boolean expression:

$$F(A,B,C) = \bar{A}\bar{B}C + \bar{A}BC + A\bar{B}\bar{C} + A\bar{B}C + ABC$$

- (i) Use Logisim to build a circuit for this function. Use only AND, OR and NOT gates. How many gates do you need?
- (ii) Write down the truth table for this function,

## Activity 2: Boolean Equation Simplification (using K-map):

Given a Boolean Expression:

$$F(X,Y,Z) = Y(\overline{X}Z + \overline{Z}) + \overline{Y}(XZ + \overline{Z})$$

- (i) Write down the truth table,
- (ii) Draw a logic circuit using the original Boolean expression,
- (iii) Simplify the equation using K-map,
- (iv) Draw the logic circuit using the simplified Boolean expression.

## Activity 3: Boolean Equations from Truth Tables

The truth table below describes a Boolean function with inputs A, B, C and D and output Z.

- (i) Create a Boolean equation using the SOP method
- (ii) Build a K-map and
- (iii) Derive a simplified equation from the K-map.

| A | B | C | D | Z |
|---|---|---|---|---|
| 0 | 0 | 0 | 0 | 1 |
| 0 | 0 | 0 | 1 | 1 |
| 0 | 0 | 1 | 0 | 1 |
| 0 | 0 | 1 | 1 | 0 |
| 0 | 1 | 0 | 0 | 1 |
| 0 | 1 | 0 | 1 | 1 |
| 0 | 1 | 1 | 0 | 0 |
| 0 | 1 | 1 | 1 | 1 |
| 1 | 0 | 0 | 0 | 0 |
| 1 | 0 | 0 | 1 | 0 |
| 1 | 0 | 1 | 0 | 1 |
| 1 | 0 | 1 | 1 | 0 |
| 1 | 1 | 0 | 0 | 0 |
| 1 | 1 | 0 | 1 | 0 |
| 1 | 1 | 1 | 0 | 0 |
| 1 | 1 | 1 | 1 | 1 |

## Post-class Activity: Reflection

Complete your weekly reflection activity in the Ed Lesson. You would be able to copy/paste your reflections (**more than 200 words in each week**) to meet assignment hurdle requirement.

## Assignment 1 Part 3 Practice Quiz Reminder

Don't forget to attempt the **Assignment 1 part 3 practice Quiz** in Moodle (only two attempts allowed).